Explicit Quadratic Height Bounds for Elliptic Multiples

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Abstract

For a specified point on an elliptic curve over $\mathbb{Q}(\sqrt{2})$, we prove an explicit upper height bound quadratic in its integer index. The proof uses exact duplication and differential-addition identities, an elementary induction, and a comparison between absolute height and primitive coefficient height. We also bound the coefficients of finite products excluding these elliptic abscissas. The results are unconditional and require no determination of the full Mordell–Weil group.

1 Quadratic height bounds for the fixed elliptic multiples

Put $K = \mathbb{Q}(\sqrt{2})$ and

$$D = 11 - 6\sqrt{2}, \qquad E:\ y^2 = x^3 - Dx, \qquad P = \left(\frac{57 - 28\sqrt{2}}{8}, \frac{-660 + 417\sqrt{2}}{32}\right).$$

The height h below is the absolute logarithmic Weil height. If $x = (a + b\sqrt{2})/c$ with $a, b, c \in \mathbb{Z}$, $c > 0$, and $\gcd(a, b, c) = 1$, put

$$h_0(x) = \lceil \log_2 \max(1, |a|, |b|, c) \rceil .$$

Lemma 1.1. *The point P belongs to $E(K)$ and has infinite order.*

Proof. Substitution verifies the curve equation. At the unique place above two, normalize v by $v(2) = 1$. The displayed coordinates have valuations $v(x(P)) = -3$ and $v(y(P)) = -9/2$, so the formal parameter $z = -x/y$ has $v(z(P)) = 3/2$. The integral formal group of this Weierstrass equation is odd: for every integer m, $[m](Z) = mZ + Z^3 g_m(Z^2)$ with integral coefficients. At a parameter of valuation at least $3/2$, multiplication by an odd integer preserves its valuation, and multiplication by two raises it by one. In each case the leading term has strictly smaller valuation than the remaining terms. Thus

$$v(z([n]P)) = 3/2 + v_2(n) \qquad (n \neq 0).$$

It is finite, and hence $[n]P \neq \mathcal{O}$ for every nonzero n. In particular all its abscissas are finite and nonzero. □

Lemma 1.2. *For $n \geq 1$, write $h_n = h(x([n]P))$. Then*

$$h_{2n} \leq 4h_n + \log 392,$$
$$h_{2n+1} \leq 2h_n + 2h_{n+1} + \log 8036.$$

The second formula is used for $n \geq 1$.

Proof. The duplication and differential-addition identities are

$$x(2Q) = \frac{(x(Q)^2 + D)^2}{4x(Q)(x(Q)^2 - D)}, \qquad x(Q+R)x(Q-R) = \left(\frac{x(Q)x(R) + D}{x(Q) - x(R)}\right)^2.$$

For the second identity, the usual addition formulas reduce it to

$$(u+v)^2(uv-D)^2 - 4uv(u^2-D)(v^2-D) = (u-v)^2(uv+D)^2,$$

which is a polynomial identity. Taking $Q = [n]P$ and $R = [n+1]P$ makes $x(Q-R) = x(P)$. Its denominator is nonzero: an equality $x([n]P) = x([n+1]P)$ would contradict the infinite order of P.

For each place, bound a homogeneous polynomial by the largest coordinate norm to its degree. At a finite place the coefficient bound is the maximum coefficient norm; at an infinite place use the sum of the coefficient norms. Applied to duplication, whose homogeneous numerator and denominator have degree four, this gives

$$h(x(2Q)) \le 4h(x(Q)) + 2h(D) + \log 8.$$

Applied to differential addition, with bidegree $(2,2)$ in the two abscissas and denominator $x(P)(x(Q) - x(R))^2$, it gives

$$h(x(Q+R)) \le 2h(x(Q)) + 2h(x(R)) + 2h(D) + h(x(P)) + \log 4.$$

For example, the archimedean coefficient sums for the second formula are bounded by $4\max(1,|D|)^2\max(1,|x(P)|)$, and the corresponding finite-place bound has no factor four. These local estimates sum to the stated absolute-height estimates.

Both real conjugates of D exceed one, and $N_{K/\mathbb{Q}}(D) = 49$, so $h(D) = \log 7$. Both real conjugates of $x(P)$ exceed one, its norm is $41^2/8^2$, and its denominator ideal has norm 8^2. Consequently $h(x(P)) = \log 41$. Substitution gives $2h(D) + \log 8 = \log 392$ and $2h(D) + h(x(P)) + \log 4 = \log 8036$. □

Theorem 1.3. *For every integer $n \ge 1$,*

$$h(x([n]P)) \le n^2 \log 23534 - \log 574, \qquad h_0(x([n]P)) \le 30n^2.$$

These inequalities are unconditional for the displayed curve and point.

Proof. Write $A = \log 23534$ and $B = \log 574$. The integer identities

$$23534 = 41 \cdot 574, \qquad 574^3 = 23534 \cdot 8036, \qquad 574^3 \ge 392$$

give $h_1 = A - B$, $3B = A + \log 8036$, and $3B \ge \log 392$. Induction using Lemma 1.2 proves $h_n \le An^2 - B$. For an even index,

$$h_{2n} \le 4(An^2 - B) + \log 392 \le A(2n)^2 - B.$$

For an odd index at least three,

$$\begin{aligned} h_{2n+1} &\le 2(An^2 - B) + 2(A(n+1)^2 - B) + \log 8036 \\ &= A(2n+1)^2 + A - 4B + \log 8036 = A(2n+1)^2 - B. \end{aligned}$$

We next compare coefficient height with absolute height. Write $x = (a + b\sqrt{2})/c$ in the primitive form above, and let $\mathfrak{d} = \{u \in \mathcal{O}_K : ux \in \mathcal{O}_K\}$ be its denominator ideal. Since $\mathcal{O}_K = \mathbb{Z}[\sqrt{2}]$, the integer c is the least positive rational integer with $cx \in \mathcal{O}_K$. At a prime above a rational prime p, use the integer-valued ideal valuation $\mathrm{ord}_{\mathfrak{p}}$, so that $\mathrm{ord}_{\mathfrak{p}}(p) = e_{\mathfrak{p}}$, and put

$d_\mathfrak{p} = \max(0, -\operatorname{ord}_\mathfrak{p}(x))$. The p-adic exponent of $N\mathfrak{d}$ is $\sum_{\mathfrak{p}|p} f_\mathfrak{p} d_\mathfrak{p}$. Multiplying by a ramification index $e_\mathfrak{p}$ makes this at least $d_\mathfrak{p}$. Therefore $(N\mathfrak{d})x$ is integral and $c \mid N\mathfrak{d}$. If $h_{\mathrm{fin}}(x)$ denotes the finite-place part of $h(x)$, then $N\mathfrak{d} = \exp(2h_{\mathrm{fin}}(x))$.

Writing x' for the real conjugate, one has

$$|a| = c\left|\frac{x+x'}{2}\right| \le c\max(|x|,|x'|), \qquad |b| = c\left|\frac{x-x'}{2\sqrt{2}}\right| \le c\max(|x|,|x'|).$$

The product $\max(1,|x|)\max(1,|x'|)$ equals $\exp(2h_\infty(x))$. It follows that

$$\max(1,|a|,|b|,c) \le \exp(2h(x)).$$

Finally $23534 < 2^{15}$, so the first height bound gives $\exp(2h_n) < 2^{30n^2}$ and hence $h_0(x([n]P)) \le 30n^2$. □

1.1 The indexed exclusion polynomial

Let $\mathcal{A}$ be a finite free algebra over an integer polynomial ring, with a fixed basis. For a multiplication matrix use the maximum column sum of the coefficient ℓ^1-norms. This norm is submultiplicative. Suppose $b \ge 0$ and the multiplication matrices of $N_0, N_1, D_0 \in \mathcal{A}$ have norm at most 2^b. Write $x([n]P) = (a_n + b_n\sqrt{2})/c_n$ in primitive form and set

$$\mathcal{E}_n = (c_nN_0 - a_nD_0)^2 - 2(c_nN_1 - b_nD_0)^2, \qquad \mathcal{E}_{\le M} = \prod_{n=1}^{M} \mathcal{E}_n.$$

Each fixed triple is computed by exact elliptic arithmetic. The product is finite and is a literal polynomial specification, independently of whether its coefficients are expanded.

Proposition 1.4. *For every integer $M \ge 1$,*

$$\|M_{\mathcal{E}_{\le M}}\| \le 2^{20M^3+30M^2+(2b+14)M} \le 2^{(2b+64)M^3}.$$

If $b \le 1048$ and $M \ge 64$, the first exponent is at most $21M^3$.

Proof. Theorem 1.3 and the triangle inequality give

$$\|M_{c_nN_0-a_nD_0}\|,\ \|M_{c_nN_1-b_nD_0}\| \le 2^{30n^2+b+1}.$$

Squaring and summing the two terms of $\mathcal{E}_n$ therefore gives $\|M_{\mathcal{E}_n}\| \le 2^{60n^2+2b+4}$. Sum these logarithmic bounds and use $\sum_{n=1}^{M} n^2 = M(M+1)(2M+1)/6$. The second bound follows from $M, M^2 \le M^3$. For the last assertion, $2b + 14 \le 2110$, and

$$30 \cdot 64 + 2110 = 4030 < 64^2.$$

The function $M^2 - 30M$ is increasing for integers $M \ge 64$. Thus $30M^2 + (2b+14)M \le M^3$, proving the result. □

Under an algebra homomorphism $\mathcal{A} \to \mathbb{Q}$ with the image of D_0 nonzero, the factor $\mathcal{E}_n$ vanishes exactly when $(N_0 + N_1\sqrt{2})/D_0 = x([n]P)$. Indeed a rational pair satisfies $u^2 - 2v^2 = 0$ only when both entries vanish. The displayed product therefore gives an exact polynomial exclusion of these finitely many abscissas, with the proved coefficient bound.